

Mohammadmahdi Gheisari

AI Engineer — Computer Vision & Edge AI

Turin, Italy · Authorised to work in Italy · Open to hybrid or on-site roles
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PROFILE

AI Engineer who ships real-time computer vision to production on edge hardware. As CTO of an early-stage startup, I productionized 5 thermography defect-detection models on GPU-edge devices at the sensor's full 25 FPS (94 FPS benchmarked), cut inspection setup time by 97% and led a team of 6 engineers. M.Sc. in Data Science and Engineering from Politecnico di Torino, with a thesis on one-shot learning for thermographic vision. I also build LLM and RAG systems, from an on-prem RAG pipeline to an offline-capable LLM application.

SKILLS

Computer vision & ML	PyTorch · TensorFlow · OpenCV · scikit-learn · anomaly detection · one-shot and low-data learning · self-supervised learning · thermographic imaging
Edge & deployment	ONNX · TensorRT · NVIDIA Jetson · GPU-edge inference · Docker · Kubernetes · GitHub Actions (CI/CD) · automated testing
Data & backend	Python · C++ · SQL · PostgreSQL · ETL · Django · REST APIs · WebSockets
Generative AI	Anthropic, OpenAI and Gemini APIs · RAG · PostgreSQL/pgvector · Ollama · llama.cpp
Languages	English (C1) · Italian (A2) · Persian (native)

EXPERIENCE

Chief Technology Officer (Machine Learning & Computer Vision) — Therness S.r.l. Turin · Sep 2025 – Jan 2026 (early-stage startup)

- Productionized 5 defect-detection models (PyTorch to ONNX/TensorRT) on GPU-edge devices for industrial thermographic inspection.
- Ran inference at the sensor's full 25 FPS capture rate in production; benchmarked at 94 FPS unconstrained (~3.8× headroom), showing the bottleneck was camera capture and stream handling, not the model.
- Cut inspection setup time by 97% (33 minutes to under 53 seconds) by re-engineering the workflow with Django and WebSockets.
- Led a 6-engineer team from R&D prototypes to production-grade systems and owned customer-facing technical delivery.
- Architected an on-prem RAG pipeline for SOP retrieval (Ollama, PostgreSQL/pgvector) with citation-enforced answers and no customer data leaving the client's environment.

R&D Scientist — Computer Vision & Deep Learning — Therness S.r.l. Turin · Sep 2024 – Aug 2025

- Developed the first thermography-based defect-detection prototypes, using one-shot and low-data learning to work with very few labeled samples.
- Built data-validation, preprocessing and error-analysis workflows to benchmark models on real-world thermal video runs.

Data Scientist Intern — Ades Consulting & Service S.r.l. Brescia · Jul 2024 – Sep 2024

- Built a robotic weld path-planning pipeline from STL meshes, matching ~87% of reference trajectories; wrote Python tooling that shortened stakeholder review cycles.

PROJECTS

NeuroLingo — offline-capable LLM application (independent) Turin · Jan 2026 – Present

- Built a provider-agnostic LLM routing layer (Anthropic, OpenAI, Gemini) with failover to a local GGUF model via llama.cpp, so the app works with no network and no API key.
- Implemented RAG from scratch for mobile size limits (hashing-trick embeddings over a NumPy vector store); 212 passing tests, packaged cross-platform with Flet/Flutter and PyInstaller.

EDUCATION

M.Sc. in Data Science and Engineering — Politecnico di Torino Turin · Mar 2026

Thesis: One-Shot Learning algorithms in computer vision tasks on thermographic acquisition

B.Sc. in Applied Statistics — University of Hormozgan Bandar Abbas, Iran · May 2017

Certifications: Machine Learning & Deep Learning Specialization (DeepLearning.AI) · Agile & Scrum (IBM)